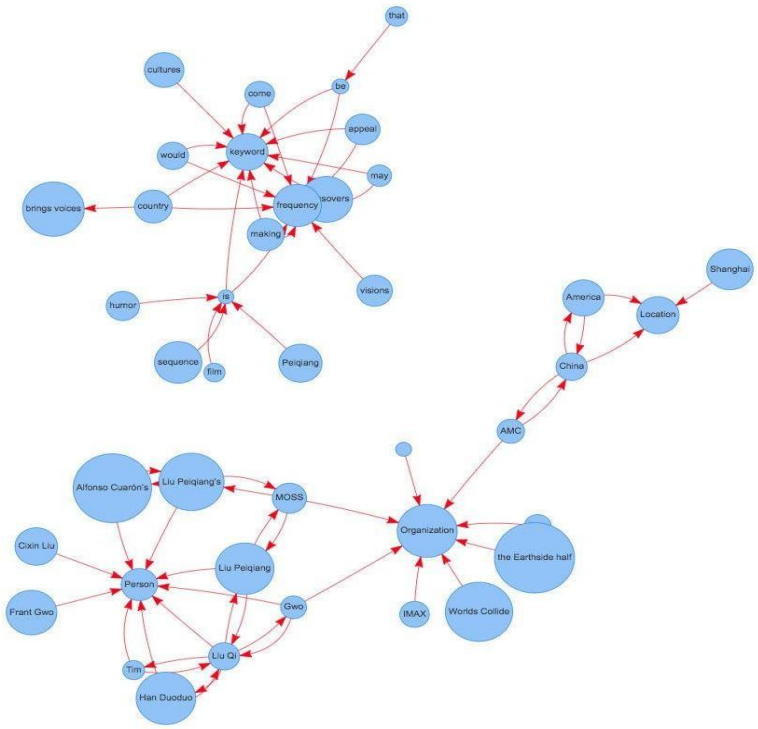
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** TextRank for Document Summarization	
Experiment No: 10	**Date:**	**Enrolment No: 92410133012**

Aim: TextRank for Document Summarization

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.

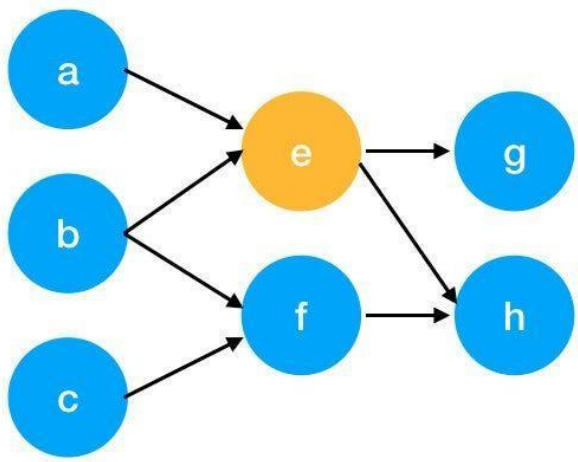


Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$In(v_e) = \{a, b\}, j \in \{a, b\}$$

$$\begin{aligned}
\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
&= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
&= S(V_a) + \frac{1}{2} S(V_b)
\end{aligned}$$

We can get the weight of webpage e by the following function.

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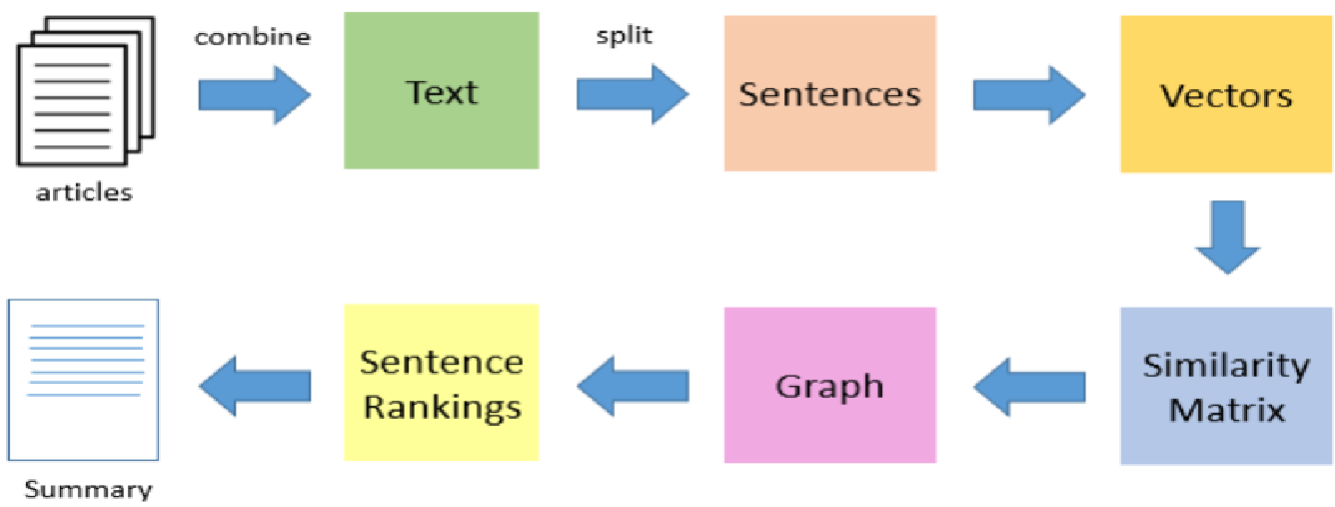
$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

TextRank for document Summarization



TextRank works in the following steps:
1. Tokenize documents into sentences.
2. Preprocess each sentence in the document.
3. Count key phrases and normalize them or produce TFIDF Matrix, you can also use any kind of vectorization such as spacy vectors.
4. Calculate the Jaccard Similarity between sentences and key phrases.
5. Rank the sentences with higher significance.

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Pre Lab Exercise:

1. **How keyword extraction is similar to document summarization process?**

- Both try to find **important information** from a text
- Both use **text processing techniques** like tokenization and removing stopwords
- Both use **ranking methods** (like TextRank)
- Both reduce the size of original text

In short: *Both focus on extracting the most meaningful parts of a document.*

2. **How keyword extraction is different to document summarization process?**

Keyword Extraction	**Document Summarization**
Extracts only **important words/phrases**	Extracts **full sentences**
Output is **keywords list**	Output is **summary paragraph**
Smaller output	Larger than keywords
Focus on **topics**	Focus on **meaning/context**

3. **Limitation of TextRank as a summarizer**

- Does not understand **actual meaning**, only similarity
- Sometimes selects **repeated or similar sentences**
- Ignores **grammar flow**
- May miss **important sentence**
- Works poorly on **very small or very large text**

Program (Code):

To be attached with

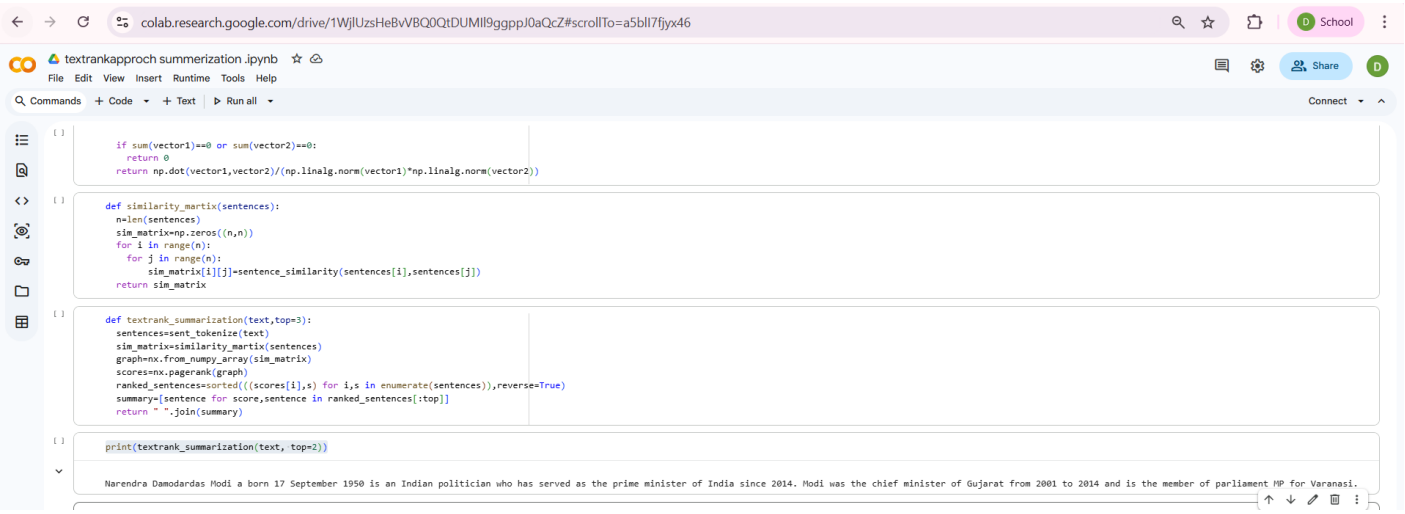
```
# Import libraries
import re
import numpy as np
import networkx as nx
import nltk

nltk.download('punkt')
nltk.download('stopwords')

from nltk.tokenize import sent_tokenize, word_tokenize
from nltk.corpus import stopwords
```

```
# Input text
text = """
Narendra Damodardas Modi a born 17 September 1950 is an Indian politician who has served as the prime minister of India since 2014. Modi was the chief minister of Gujarat from 2001 to 2014 and is the member of parliament MP for Varanasi. He is a member of the Bharatiya Janata Party BJP and of the Rashtriya Swayamsevak Sangh (RSS), a right-wing Hindutva paramilitary volunteer organisation. He is the longest-serving prime minister outside the Indian National Congress.
"""

# Preprocessing
def preprocess_sentence(sentence):
    stop_words = set(stopwords.words("english"))
    word_tokens = word_tokenize(sentence.lower())
    words = [w for w in word_tokens if w.isalnum() and w not in stop_words]
    return words

# Sentence similarity
def sentence_similarity(sent1, sent2):
    words1 = preprocess_sentence(sent1)
    words2 = preprocess_sentence(sent2)

    all_words = list(set(words1 + words2))

    vector1 = [words1.count(w) for w in all_words]
    vector2 = [words2.count(w) for w in all_words]

    if sum(vector1) == 0 or sum(vector2) == 0:
        return 0

    return np.dot(vector1, vector2) / (np.linalg.norm(vector1) * np.linalg.norm(vector2))

# Similarity matrix
def similarity_matrix(sentences):
    n = len(sentences)
    sim_matrix = np.zeros((n, n))

    for i in range(n):
        for j in range(n):
            if i != j:
                sim_matrix[i][j] = sentence_similarity(sentences[i], sentences[j])

    return sim_matrix

# TextRank summarization
def textrank_summarization(text, top=2):
    sentences = sent_tokenize(text)

    sim_matrix = similarity_matrix(sentences)
```

```
    graph = nx.from_numpy_array(sim_matrix)
    scores = nx.pagerank(graph)

    ranked_sentences = sorted(((scores[i], s) for i, s in enumerate(sentences)), reverse=True)

    summary = [sentence for score, sentence in ranked_sentences[:top]]
    return " ".join(summary)

# Output
print(textrank_summarization(text, top=2))
```

Results:

To be attached with



Observation:

- TextRank selects sentences based on **importance score**
- It uses **similarity + graph ranking**
- Output is **short and meaningful**
- But sometimes not perfectly human-like

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Post Lab Exercise:

1. Write about "your ownself" (in not more than 500 words You know better about you, rather than ChatGPT!!). Generate the summary of your portfolio in
   a. 50% of the size of your portfolio
   b. 25% of the size of your portfolio

"I am Priyanshi Maru, a second-year computer engineering student. I completed my diploma in 2024 and now I am studying at Marwadi University. I am interested in software development, AI, and cloud computing. I have worked on projects like smart attendance system with emotion detection and blood group detection using image processing. I also have knowledge of Python, SQL, and web development. I enjoy learning new technologies and improving my skills. My goal is to become a software engineer in a top company like Google or Amazon. I also love traveling, music, and spending time with my family."

Colab output :-

```
[1] 0s
print(textrank_summarization(text, top=2))

I also have knowledge of Python, SQL, and web development. I am interested in software development, AI, and cloud computing.
```

a) 50% Summary
"I am Priyanshi Maru, a computer engineering student at Marwadi University. I have worked on projects like smart attendance system and blood group detection. I know Python, SQL, and web development. My goal is to become a software engineer in top companies."

b) 25% Summary
"Priyanshi Maru is a computer engineering student skilled in Python and web development, aiming to become a software engineer."

Comment over the summary obtained. Which sentence you think should be there in your summary, but was not spotted by textrank? Rate the overall summary in the scale of 1-5 (with 1 as least and 5 as highest rating). Paste the code, your portfolio, output and your analysis.

Comment on Summary

- Summary is **clear and short**
- Some personal details like **hobbies missing**
- Important line missing:
- *"I love traveling and music"*

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